ANALYSIS OF A COMPLEX IRON SALT

OBJECTIVES

For you, the student, to write….

SAFETY PRECAUTIONS

Safety goggles, aprons, and gloves must be worn at all times in the laboratory. H_2SO_4 and concentrated H_3PO_4 are corrosive acids; wash all affected areas thoroughly with cold water. Acetone and alcohol are flammable reagents; extinguish any open flames or spark sources in the lab. Sodium oxalate is very toxic via oral and inhalation routes and severe kidney damage is possible if oxalate salts are taken internally. Oxalate compounds can be absorbed through the skin; wear gloves and wash affected areas with cold water. $KMnO_4$ is a very strong oxidizing agent; <u>do NOT pour any permanganate solutions into the ORGANIC collection bottles.</u> Permanganate solutions can stain skin and clothing.

INTRODUCTION

The complex iron salt prepared in a previous experiment is known to contain potassium, iron (III), oxalate, and loosely bound waters of hydration. The general formula is: $K_xFe_y(C_2O_4)_z \cdot n\,H_2O$. The mass percent oxalate ion in the salt will be determined by titration with a standardized $KMnO_4$ solution as shown in the <u>unbalanced</u> reaction below:

$$(1) \qquad MnO_4^{-}(aq) + C_2O_4^{2-}(aq) \rightarrow Mn^{2+}(aq) + CO_2(g)$$

Since aqueous solutions of permanganate ion are not stable over a long period of time, the exact concentration of $KMnO_4$ must be determined by titration with a known amount of a primary standard salt such as sodium oxalate, $Na_2C_2O_4$. After $KMnO_4$ has been standardized, the complex iron salt can be titrated to determine its oxalate content. The solutions containing $C_2O_4^{2-}$ and Mn^{2+} ion are colorless; the MnO_4^- solution is a deep purple color. Therefore, the titrated solution will remain colorless until all the oxalate salt is consumed in the reaction. The endpoint corresponds to the appearance of the first permanent pink color due to the presence of excess unreacted permanganate ion. The rate of the reaction is very slow at room temperature so the solution must be heated to 80°C to observe the color change in "real time". Often there is a 30-60 second time lag at the beginning of the titration before the color changes begin to take effect because the reaction takes place in a series of steps and an intermediate must be formed before the reaction goes to completion.

Interferences can occur in redox titrations from impurities in the solvent which act as reducing or oxidizing agents. Corrections are easily made by titrating a "blank" containing only the solvent. The "corrected volume" is equal to the volume of $KMnO_4^-$ required to titrate oxalate ion in solvent minus the volume required to titrate the solvent alone. (Important note: The amount needed to titrate the blank is often only one or two drops of $KMnO_4^-$.)

Another interference in the redox titration is caused by Fe(III) ion. Fe^{3+} is released into solution when permanganate ion reacts with oxalate ion and destroys the complex ion. Ferric ion is reddish colored and can interfere with observation of the faint pink titration endpoint. To eliminate the color interference, a small amount of concentrated phosphoric acid is added to the solution. The phosphate ion reacts with Fe^{3+} to yield a colorless complex ion, $Fe(PO_4)_2^{3-}$, and the reddish-brown color of Fe^{3+} is eliminated from solution.

To determine the mass percent of iron, Fe(III) must first be reduced to Fe(II) by exposure to sunlight or by reaction with Al metal. The resulting Fe^{2+} ion is then titrated with a standardized $KMnO_4$ solution according to the <u>unbalanced</u> equation below:

$$(2) \qquad Fe^{2+}(aq) + MnO_4^-(aq) \rightarrow Fe^{3+}(aq) + Mn^{2+}(aq)$$

The mass percent of water is determined by gravimetric analysis. A known mass of the complex salt containing water is weighed, heated and reweighed. The weight of water is the mass difference between the hydrous and anhydrous forms of the salt.

The mass percent of potassium in the salt is determined by difference using the experimentally determined masses of iron, oxalate and water and the mass of the complex iron salt.

PROCEDURE

(Note: If you were not able to synthesize the complex iron salt in a previous experiment you should skip part A and go to part C. Borrow crystals from a labmate.)

Part A: Preparation of the Dried Salt

Use vacuum filtration to separate the crystals from solution. Wash the green crystals in the Buchner funnel with small quantities of 1:1 (by volume) ethanol and water solution and follow with small amounts of acetone to remove impurities and excess oxalate ion. Dry the crystals by continuing to draw air through the filter cake in the funnel for about 10 minutes. Pour the green alcohol/acetone filtrate solution into the waste bottles. After taking the mass of the dried green crystals, transfer them to a scintillation vial wrapped with Al foil (to avoid slow decomposition of the complex iron salt caused by exposure to light). Label the scintillation vial with your name, date, contents ($K_xFe_y(C_2O_4)_z \cdot nH_2O$), and mass.

Standardization of $KMnO_4$ Solution

<u>This part will be done by the Stockroom</u>. Make sure you record the exact concentration of $KMnO_4$ before leaving lab; you will need it for your post-lab calculations. The procedures followed by the stockroom personnel are:

An acidic solution of sodium oxalate ($Na_2C_2O_4$) was prepared by dissolving approximately 0.100 g of $Na_2C_2O_4$ in 150 mL of 1.0 M H_2SO_4 in a 250 mL Erlenmeyer flask. The resulting solution was heated to 80°C and then titrated with the $KMnO_4$ solution until the first appearance of a permanent faint pink color. A "blank" containing 150 mL of 1.0 M H_2SO_4 was also titrated. After titration was completed, the titrated solutions and remaining $KMnO_4$ were neutralized according to the instructions in Part D.

Part C: Analysis of Oxalate Ion in the Complex Iron Salt

(Do not begin part C until all the green filtrate solutions from Part A have been removed from the lab by the stockroom staff.) Weigh approximately 0.1 g of the complex iron salt and record its mass to the milligram. Dissolve the salt in 10-mL of 6 M H_2SO_4 in a small beaker. Pour the acid solution carefully into 75 mL of DI water in a 250 mL Erlenmeyer flask and add 1-mL of concentrated H_3PO_4. Heat the solution to 80°C, stirring gently with a magnetic stir bar. Titrate the heated solution with $KMnO_4$ until the appearance of the first permanent faint pink color, recording the initial and final volumes of $KMnO_4$ used. *Repeat this titration at 2-3 more times.* Prepare a blank using the same volumes of DI water, 6M H_2SO_4 and concentrated H_3PO_4 which are used to titrate the iron salt, titrate. When finished, pour all titrated solutions and any $KMnO_4$ remaining in your buret into a 1000 mL beaker. Rinse the buret and buret tip with a small amount of DI water and add the rinse water to the beaker. No pink color should be left in the buret when finished.

Part D: Reducing the Solution

IN THE HOOD, slowly mix sodium bisulfite ($NaHSO_3$) solution into the beaker containing the titrated solutions and leftover $KMnO_4$ until the mixture turns clear and colorless (3). Do not go on to the next step until the solution is completely colorless!

$$(3) \qquad 2MnO_4^{-}(aq) + 6H^{+}(aq) + 5SO_3^{2-}(aq) \rightarrow 2Mn^{2+}(aq) + 5SO_4^{2-}(aq) + 3H_2O(l)$$

Part E: Neutralizing the Solution

The colorless mixture will be very acidic. To neutralize it, place about ½ inch of solid $NaHCO_3$ in the bottom of a clean 1000 mL beaker, add water until it barely covers the $NaHCO_3$, stir to make a slurry. Now slowly pour the solution from the first 1000 mL beaker into the bicarbonate slurry and stir. When foaming stops, check the solution with pH paper. If pH = 7, pour the neutralized solution into a collection bottle in the hood. If the pH of the solution is less than 7, prepare a new slurry and neutralize the solution again in the same manner.

(4) $H^+(aq) + NaHCO_3(s) \rightarrow Na^+(aq) + H_2O(l) + CO_2(g)$

Be sure to clean your lab table, glassware and buret thoroughly before leaving lab. Return the test tube containing unused crystals to the TA before leaving lab.

Before coming to lab prepare your duplicate page notebook as before:

Your name, ID#, TA name, and date should appear in the upper right-hand corner or in the space provided for this information on each page. The experiment title and reference source should appear at the top of the first page, followed by a brief statement of objectives, chemical and equipment tables, and then the procedures and observations. The calculations below should follow the procedures and observations.

CALCULATIONS

(1) Using the standardized concentration of $KMnO_4$ and your own data from Part C, calculate the mass percent of oxalate ion in the complex iron salt. For the calculations below, you may average the mass percents or use the result that gives the most reasonable result.

(2) Because of time constraints, the following data has been collected for you. In a previous experiment, an empty crucible was heated to drive off any volatile impurities and weighed after cooling (see below). A small amount of the green complex iron salt was added to the crucible, weighed again, and then heated gently to remove any waters of hydration loosely bound to the salt. The crucible containing the anhydrous iron salt was weighed a final time after cooling. Find the mass percent of water in the complex iron salt.

mass of empty crucible:	12.543 g
mass of crucible and salt before heating:	12.670 g
mass of crucible and salt after heating:	12.656 g

(3) Because of time constraints, the following data has been collected for you. In a previous experiment, 0.800 g of the green complex iron salt was placed in an Erlenmeyer flask and heated with concentrated sulfuric acid until all oxalate ion was destroyed (CO_2, SO_2 and H_2O are formed). Ferric ion (Fe^{3+}) remained in solution and was reduced to Fe^{2+} by reaction with Al wire. The ferrous ion (Fe^{2+}) was then titrated with $KMnO_4$ solution according to equation (2) and 32.60 mL of 0.0100 M permanganate solution was required to reach the endpoint. The blank required 0.03 mL of the permanganate solution. Find the mass % of iron in the complex salt.

(4) With the mass percent of Fe, oxalate ion, and water (calculated above); find the mass percent of potassium ion in the complex iron salt by difference.

(5) Find the empirical formula of the complex iron salt, $K_xFe_y(C_2O_4)_z{\cdot}nH_2O$ using the mass percent of potassium, iron, oxalate and water.

(6) Insert your values for x, y, z, & n into $K_xFe_y(C_2O_4)_z{\cdot}nH_2O$ and provide the correct coefficients to balance the following equation:

$$\_\_\_\ Fe(NH_4)_2(SO_4)_2{\cdot}6H_2O + \_\_\_\ H_2C_2O_4 + \_\_\_\ K_2C_2O_4 + \_\_\_\ H_2O_2$$
$$\rightarrow \_\_\_\ K_xFe_y(C_2O_4)_z{\cdot}nH_2O + \_\_\_\ (NH_4)_2SO_4 + \_\_\_\ H_2SO_4 + \_\_\_\ H_2O$$

(*Hint:* Remember H_2O_2, an oxidizing agent, was used in the synthesis, so you'll need to do a redox balance. Start with just the iron ions, balance the reaction, and then add the full formulas.)

(7) Using the balanced equation from #6 and the mass of $Fe(NH_4)_2(SO_4)_2{\cdot}6H_2O$ recorded at the beginning of last week's experiment determine the percent yield of $K_xFe_y(C_2O_4)_z{\cdot}nH_2O$.